



Nebraska Public Power District

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10 CFR 50.73

NLS2023028

May 11, 2023

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2022-004-01
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2022-004-01.

There are no regulatory commitments contained in this letter.

Sincerely,

Khalil Dia
Site Vice President

/jo

Attachment: Licensee Event Report 2022-004-01

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

NRC FORM 366 COMMISSION (03-14-2023)		U.S. NUCLEAR REGULATORY LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	
APPROVED BY OMB: NO. 3150-0104 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: aira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.		EXPIRES: 08/31/2023	

1. Facility Name Cooper Nuclear Station		☒ 050 ☐ 052	2. Docket Number 00298	3. Page 1 of 4
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4. Title Manual Reactor Scram and Group I Isolation due to Main Turbine Bypass Valve Failing Open				
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
12	16	2022	2022	- 004 -	01	05	11	2023		050
									Facility Name	052 Docket Number

9. Operating Mode	10. Power Level
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)					
10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER	
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager	Phone Number (Include area code) (402) 825-5416

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	SO	CL		Y					

14. Supplemental Report Expected			15. Expected Submission Date		
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)		Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)
A reduction in power was performed on December 15, 2022, to repair a hydrogen leak on the main generator. On December 16, Operations personnel noted that both electro-hydraulic (EH) oil pressure and reactor vessel pressure were lowering and that two of three turbine bypass valves were closed with one full open. Operations personnel responded by starting a second EH fluid pump to restore EH system pressure, however, reactor vessel pressure continued to lower. As a result of reactor vessel pressure lowering, a manual reactor scram was inserted. Immediately following the manual reactor scram, a Primary Containment Group 1 Isolation was received. The cause of one of the turbine bypass valves being open was the failure of the fast-open solenoid operated valve (SOV) coil. All three of the SOV coils were sent to a lab for analysis to determine if there were any additional failure modes of the coil. The analysis determined the most likely cause of the premature failure of the SOV was a shorting of the coil due to a weak point along the coil wiring breaking down over time. The other two SOV coils, which did not fail, were also examined and found to be in good condition with no gross flaws. There was no impact to nuclear safety, radiological safety, or industrial safety. This event was reported under Event Notification 56278 on December 17, 2022.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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FACILITY NAME Cooper Nuclear Station	X 050	2. DOCKET NUMBER 00298	3. LER NUMBER		
	052		YEAR 2022	SEQUENTIAL NUMBER - 004	REV NO. - 01

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 1, Power Operation, at approximately 13 percent power, with the main turbine generator [EIS:TA] out of service, at the time of the event on December 16, 2022.

BACKGROUND

The Main Turbine Bypass System [EIS:SO] is designed to control steam pressure when reactor steam generation exceeds turbine [EIS:TB] requirements during unit startup, sudden load reduction, and cooldown. It allows excess steam flow from the reactor [EIS:RPV] to the condenser without going through the turbine. The bypass capacity of the system is 25 percent of the Nuclear Steam Supply System [EIS:SB] rated steam flow. Sudden load reductions within the capacity of the steam bypass can be accommodated without safety relief valves [EIS:RV] opening or a reactor scram. The Main Turbine Bypass System consists of three bypass valves (BPV) [EIS:V] connected to the main steam lines between the main steam isolation valves [EIS:ISV] and the turbine stop valves.

The turbine utilizes a Digital Electro-Hydraulic (DEH) [EIS:TG] system to control reactor pressure by positioning governor valves and condenser bypass valves. It consists of solid state governor devices, governor, startup control devices, emergency devices for turbine and plant protection (overspeed governor, master trip, vacuum trip, motoring protection, thrust bearing wear trip, low bearing oil pressure trip) and special control and test devices. The control system operates the main stop valves, governor valves, bypass valves, reheat stop and intercept valves and other protective devices. DEH system oil pressure is maintained by two hydraulic pumps [EIS:P] located at the DEH reservoir tank [EIS:T].

The function of the fast open permissive solenoid operated valve (SOV) is controlled by the DEH control system to be energized and closed when reactor power is below 115 MWe with the output breaker closed, or anytime the output breaker is open. During normal operation, above 115 MWe, the SOV is de-energized and open. There is a single SOV on each of the bypass valve actuators and controlled by redundant interposing relays at the DEH control cabinet rack. The fast open permissive SOV is only installed on the bypass valves.

EVENT DESCRIPTION

On December 9, 2022, a hydrogen leak was detected on the main generator. To repair this leak, a reduction in power was performed on December 15. On December 16, the three turbine BPVs were 40 percent open for reactor pressure control. At 23:49 hours, Operations personnel noted that both the electro-hydraulic (EH) oil pressure and reactor vessel pressure were lowering and that two of the three BPVs were closed with one full open. Operations personnel initially diagnosed the cause as an EH leak and responded by starting a second EH fluid pump to restore EH system pressure, however, reactor vessel pressure continued to lower.

As a result of reactor vessel pressure continuing to lower, Operations personnel inserted a manual reactor scram at 23:51 hours. Immediately following the manual reactor scram, a Primary Containment Group 1 Isolation on Main Steam Line Low Pressure was received, and all Main Steam Isolation Valves and Steam Line Drain Valves closed. As a result, the station entered Mode 3, Hot Shutdown. The nominal setpoint for a Group 1 Isolation is 835 pounds per square inch gage (psig) main steam line pressure. The manual scram was inserted at 841 psig reactor pressure. The automatic Group 1 Isolation signal was received before scram action response to remove the mode switch from the RUN position, which would have bypassed the Group 1 isolation per Procedure 2.1.5, Reactor Scram.

While the unit was in Mode 3, repairs were completed to resolve the hydrogen leak and troubleshooting was performed to determine why one BPV failed open. Initial finding determined the cause to be the failure of the fast-open SOV (TGF-SOV-BV-1) coil.

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NARRATIVE

After removal and inspection of TGF-SOV-BV-1, it was discovered the epoxy casing around the coil was bulged indicating signs of overheating. The other two coils were removed and inspected and showed no visible signs of degradation. However, the decision was made to replace all three SOV coils on the BPVs. Post work testing was completed satisfactorily, and the BPVs were returned to service to support plant startup. The station entered Mode 2, Plant Startup, at 00:52 on December 19, and Mode 1, Power Operations, at 14:42 the same day.

BASIS FOR REPORT

The event is reportable under 10 CFR 50.73(a)(2)(iv)(A) – Any event or condition that resulted in manual or automatic actuation of any of the systems listed in 10 CFR 50.73(a)(2)(iv)(B), specifically, (a)(2)(iv)(B)(1) – Reactor protection system including: reactor scram or reactor trip; and (a)(2)(iv)(B)(2) – General containment isolation signals affecting containment isolation valves in more than one system or multiple main steam isolation valves.

This event was also reported under Event Notification 56278 on December 17, 2022.

SAFETY SIGNIFICANCE

The resultant manual shutdown of the reactor is the direct consequence of the event, including the reactor manual scram in response to the decrease in reactor pressure that was caused by a failed open main turbine bypass valve. The scram occurred with no complications, including the expected and as designed isolation of the main steam isolation valves on low reactor pressure while the reactor mode switch was in the Run position.

Investigation found that the transition to safe shutdown was not complicated by additional failures. Operators achieved safe shutdown in accordance with plant procedures. Mitigating systems required for safe shutdown remained available and responded as designed. This included the availability of emergency core cooling systems, the main condenser, and feedwater systems. There was no impact to the general safety of the public, nuclear safety, industrial safety, or radiological safety.

CAUSE

Initial validation and troubleshooting identified the cause for BPV #1 to fail open was due to the fast-open SOV coil (TGF-SOV-BV-1) failing. Subsequent lab analysis determined the most likely cause of the premature failure of the SOV was a shorting of the coil due to a weak point along the coil wiring breaking down over time. The breakdown of the coil would have caused increased resistance at the weak point. As the resistance increased, the current caused excess power dissipation and heating. The heating degraded the wire coating causing the coil to short.

The other two SOV coils, which did not fail, were also examined and were found to be in good condition with no gross flaws.

The automatic Group 1 isolation occurred due to lowering reactor pressure as a result of the failed open BPV #1. The automatic Group 1 isolation was due to a gap in symptom-based response to degrading reactor pressure; specifically, Operators initially responded to a possible EH leak prior to inserting the manual reactor scram and related response actions.

CORRECTIVE ACTIONS

All three of the fast-open SOV coils were replaced.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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FACILITY NAME	X 050 052	2. DOCKET NUMBER	3. LER NUMBER		
Cooper Nuclear Station		00298	YEAR	SEQUENTIAL NUMBER	REV NO.
			2022	- 004	- 01

NARRATIVE

As a result of the lab analysis findings, the station is developing enhanced acceptance testing to reduce the likelihood of accepting or installing a defective SOV coil.

Operations training is being conducted to reinforce symptom diagnosis and response actions.

PREVIOUS EVENTS

There have been no Licensee Event Reports submitted within the past 10 years regarding failure of bypass valves resulting in a manual or automatic scram.